A Comparative Analysis of Machine Learning Models for Stock Price Prediction Performance Evaluation and Improvement Strategiesve Analysis of Machine Learning Models for Stock Price Prediction

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Abstract

Stock price prediction has always been a big challange of research in financial markets, enabling investors and analysts to make informed decisions based on historical data. Traditional methods often struggle with the inherent non-linearities and complex patterns found in stock market data.

Here, we are looking at four models' Random forest, Logistic Regression, Support Vector Machine, Gradient Descent to predict stock price. For the purpose of this analysis I gathered data from Kaggle, it includes historical prices of Apple Inc. stock (AAPL) for the past ten years. The models' performance has been measured using accuracy, and other measures were also determined to assess the performance to the best level by using (MAE), (MSE) and (RMSE).

Hence proposed model shows a high accuracy of 86.24% compared to all the three models since the Random Forest model is variability efficient in preventing over-validation and is suitable to handle big data with high dimensions.Logistic Regression also showed strong performance with an accuracy of 85.71%, highlighting its effectiveness in modeling linear relationships in the data. The Support Vector Machine (SVM) achieved an accuracy of 84.65%, showing promise, but with room for improvement through further hyperparameter tuning or the use of non-linear kernels. The Gradient Descent model, while fundamental for optimization, yielded the lowest accuracy at 78.30%, likely due to its limitations in finding non-linear relationships.

In addition to comparing model performance, we explore several techniques to improve the accuracy of the Random Forest. These include hyperparameter tuning, feature engineering, and the application of ensemble methods. In this research, important information is made regarding the efficiency of various machine learningtechniques and key areas of improvement have been identified that can be used in the further development of research endeavours and the implementation of this approach in real-world context.

Keywords: Stock price prediction, machine learning, performance evaluation, improvement strategies.

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1. Introduction

Stock price forecasting is a very essential task in financial market helping in anticipating future stock price from past data. The better the prediction the better will be the gains, it also helps the investors to make an informed decision. Normal models Make it difficult to predict data with nonlinear relationships, a phenomenon we see in financial data. This lead to the use of more advanced Machine Learning techniques that are able to learn such complex patterns better [1].

As AI (artificial Intelligence) models generally, ML models have solid potential in improving the prediction accuracy owing to their hysteresis capacity, which maneuvering patterns that do not manifest during the traditional diagnosis stages. Zhang et al. [2] proved that neural networks outperform popular autoregressive models in terms of stock price prediction. Similar work was done by Kim and Han [3] who optimized the feature selection process using genetic algorithms for neural networks, leading to enhanced predictive performance.

However, works applied ensemble learning methods with results more accurate than other alone methods for studying the relationships between financial variables has received less attention in the last few years. Random Forest is a learning paradigm that creates decision trees and combines their results in order to increase accuracy and prevent over-learning. [4]. There have also been successful attempts at applying other ensemble methods like Gradient Boosting Machines and XGBoost to stock price prediction tasks [5]. In-vestigated four machine learning models—Random Forest, Logistic Regression, Support Vector Machine (SVM), and Gradient Descent—to determine their effectiveness when predicting stock prices. These models are unique and come with different strengths and weaknesses.

For instance, Random Forest is recognized for its excellent classification and such ability also to manage large datasets [6] with greater number of dimensions. Logistic Regression, traditionally used for the binary classification task, is a very simple and interpretable model that is nonetheless applicable to regres- sion tasks as well [7]. The strength of SVMs comes from its capacity to work with both linear and nonlinear data by employing kernel functions [8]. Gradient Descent is an optimization algorithm that plays an important role in training many machine learning models specifically for linear regression and neural networks [9]. Here, the dataset used in this specific study has its relevance based on its comprehensive nature which includes high, low, close, open, volume, and adjusted values from several years. The purpose of this study is to determine the effective method for stock price prediction and analyze suggestions to improve the performance of the models by comparing the prevalence of these three models. Agricultural pest manage- ment is vital to sustaining crop health and productivity traditional pest identification methods always face the challenge of accuracy; this is because the datasets rarely show a balanced dataset of pests, resulting in biased predictions, machine learning appli- cations for pest identification has gained interest by researchers over the recent time, specifically multi-class type techniques, overall cost-sensitive learning is largely successful as far as addressing the class imbalance problem by integrating undersampling techniques 4 to 6 to rebalance imbalanced data without losing valuable information.[23] The studies have focused more on the use of deep learning models to identify plant disease [22] However, class imbalance is still a very serious challenge.

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2. Related Work

Some literature studied the implementation of machine learning models into stock price prediction, which is utilized through different descriptions anddatasets, which led to different levels of accuracy and robustness.

Patel et al. [10] provides an extensive overview of numerous machine learning models applied from ANNs, SVMs and Random Forest to predicting stock prices. As far as the results, the ensemble methods, e.g., Random Forests, prove to be the best among all, since they are much more likely to reduce overfitting, and have more chances to generalize well, due to aggregating the predictions of multiple base estimators.

Atsalakis and Valavanis [11] had an overview of the introduction of soft computing techniques such as neuro-fuzzy systems and neural networks in stock market prediction. The strengths of such models are fairly well delimited in their review; they do well in capturing the nonlinearities engrained in financial data. They added that neurofuzzy systems can learn from background knowledge in the form of neural networks, resulting in a higher accuracy in their predictions by the model than standard methods.

Chong et al. [12] Explored the deep learning models, specifically, (CNNs) and (LSTM) networks for analysis and prediction of stock market. The authors showed that deep learning motifs have learned how to shift time-series data in ways that are not possible for traditional machine learning motifs. LSTM networks were also found to outperform in predictive tasks involving stock prices.

Besides these studies, more researchers have learned on hybrid models where various machines learning models are included to compensate for weaknesses of all approaches. In particular, Fischer and Krauss [13] combine LSTM networks and a Random Forest classifier in a hybrid model that manages to predict stock prices more accurately by preserving the memory and temporal dependencies of the data as well as exploring complex interactions among the stock features.

Particularly ensemble learning methods have shown remarkable performance in forecasting financial markets. [4] Breiman's pioneering paper on Random Forests illustrates the effectiveness of ensemble learning and gave rise to other models in this area based on decision trees and also became one of the most widely used prediction techniques. More recent studies, like Nguyen et al. [5], classifiy enhanced ensemble approaches such as Gradient Boosting Machines (GBM) and Extreme Gradient Boosting (XGBoost) compared to random forest and reveal a significant difference in prediction accuracy and computational performance.

Additionally, the financial prediction with support vector machines (SVMs) has been well-known. The pioneering work of Cortes and Vapnik [8] introduced SVMs, demonstrating their ability to efficiently manage high-dimensional spaces and deliver strong prediction performance from relatively small-sized samples. Such studies (e.g. Huang et al. wrote to predict stock market using SVMs, with signals which will show the accuracy of classification and trends of data.

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Nonetheless, challenges persist, including model interpretability, data preprocessing, and overfitting. One of the major concerns here with regards to financial data modeling will be overfitting, since there will be a lot of noise and volatility in stock prices. Various techniques are generally used to avoid this problem like cross-validation, regularization, and ensemble methods, etc. Furthermore, the preprocessing steps applied to the data set, like normalization, imputation of missing values and feature engineering, are key to the performance of machine learning algorithms.

3. Proposed Methodology

In this research, we shall seek to build four models and compare their capacity to forecast stock prices; Random Forest, Logistic Regression, SVM and Gradient Descent. The following are the steps that need to be undertaken during the methodology; Collecting the data, Preprocessing of data, Training model and at last evaluating model. The detailed discussion of each step is provided in the subsequent sections.

3.1 Data Collection

The dataset was collected from the Nasdaq stock exchange, covering over ten years of daily trading data for Apple Inc. (AAPL) from January 2010 to the present. The data has main financial metrics such as high, low, opening and closing prices, trading volume and adjusted closing prices. Automated scripts were used to extract and preprocess the data, ensuring consistency and accuracy by handling missing values and adjusting for corporate actions. The dataset is updated regularly to maintain its relevance, providing a solid foundation for developing stock price forecasting algorithms.

3.2 Data Preprocessing

Predictive model preparation is a necessary step to undertake on dataset. for machine learning models. This pre- processing steps are performed:

- **Data Cleaning**: It has identified the missing values and has managed accordingly. It drops rows with missing values if the count of missing observations is less. In more significant data, the imputation used mean or median.

- **Feature Selection**: Only those features which are high, low, open, close, and volume were only considered for developing the model, adjusted close feature was used on the target.

- **Date Handling**:Dates were converted to a usable format and then split into separate features, such as year, month, and day. Such splitting of dates enabled models to capture temporal patterns.

- **Normalization/Standardization:** Some of the features were scaled between 0 and 1. Others have been rescaled that their standard deviation is = 1. It is because preprocessing was required until we get the final output result as 1 or 0 in such models like SVM and Gradient Descent.

- **Train-Test Split**:In this research, We shall come up with four models and try to

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figure out which is best predicting the stock prices; Random Forest, Logistic Regression, SVM and Gradient Descent in our study. These are the procedure that are to be carried during the methodology; Data Collection, Data preprocessing, training of the model and finally evaluation of the model. Each step of it elaborately explained in the followingsections

3.3 Model Training

From the bunch of machine learning models, the fol- lowing models are selected for this research: Random Forest, Lo- gistic regression, SVM and Gradient descent. All four models are trained on the preprocessed dataset with the training dataset of 1972.

- **Random Forest:** Random forest is a ensemble learning technique that assembles many trees during training and uses the aggregate in classification or regression mode of the classes or mean predictions of the individual trees. [4] ...We did fit the Random Forest model under several hyperparams (n estimators, max depth, min samples split).

- **Logistic Regression:** One most important characteristic of Logistic Regression is that it is a classification algorithm that can only handle with two classes; although it can be used for regression analysis. They claim it is a technique that estimates a binary dependent variable following a logistic transformation [7]. In this paper, to which the authors proposed the original model during the time-period given for both regression and adjustment, some modifications were made so that it was suitable to use in the stock price prediction task.

- **Support Vector Machine (SVM):** Besides, in this respect the Support Vector Machine (SVM) is treated as the classification algorithm that simplifies data analysis for classification and regression testing, and determines that hyperplane that provides the optimal splits of a given dataset into classes [8]. Additionally, for regression tasks, the SVM model with a linear kernel was analyzed in order to capture the relationships between features.

- **Gradient Descen**t: Gradient descent is a method of optimization to minimize the loss function in several machine learning based models [9]. In this experiment, it was applied to fine tune the weights of linear regression model. Learning rate and iterations were among the most common hyperparameters that were tuned while training.

3.4 Model Evaluation

The performance of each of the models was examined using the testing set. Accuracy is the ratio of the absolute number of actual stock price forecasted to the total number of observations which was the broadest evaluation criterion. Then each model was compared to find best effective model and completeness was assessed of each.

Out of which, Mean Absolute Error, Mean Squared Error and Root Mean Squared Error were used to compare all the models more closely. but these will only give us an average of how big the errors made in the predictions were and so give us a better way of making comparisons over how good the models are.

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3.4.1 Feature Importance Analysis

Regarding Random Forest model, the feature importance was conducted to see how each feature contributed to theprediction task. This analysis assists in selecting the relevant features that have impact in the result, and reveals thepatterns hidden in the data.

3.4.2 Improving Model Performance

Several strategies were employed to improve the performance of the Random Forest model, given its superior accuracy. These strategies include:

- **Hyperparameter Tuning:** TThere is always an important optimization of this Random Forest model through hyperparameter tuning. Some of the hyperparameters included: number of timber (n estimators), maximum depth (max depth), the minimum number of instances required to split samples in an internal node (min samples split), the minimum number of samples required to be at a leaf node (min samples leaf), and the maximum number of features to consider at each splitting time (max features) were tuned through the help of GridSearchCV and RandomizedSearchCV.

- **Feature Engineering:** Developing new features or transform existing ones co provide more information to the model. The other analytics tool used was; Moving Averages, Bollinger Bands, Relative Strength Index (RSI), lagged features using past values of the stock prices, and interaction features created by combining existing features.

- **Data Augmentation:** Increasing the size of the training set Synthetic data for data generation, or bootstrapping, can provide the model with additional examples to learn from.

- **Reducing Overfitting:** Pruning trees by setting constraints like max depth, min samples split, and min samples leaf can help reduce overfitting. Ensemble techniques like Bagging or Boosting were also considered.

- **Cross-Validation:** K-fold cross-section validation was performed, for final validation with hopes of getting the same model results consistently on different subsets of data. This technique tends to be very useful in determining the transferability of the model.

- **Handling Imbalanced Data:** Measures like oversampling of minority class, Synthetic Minority Over-sampling Technique (SMOTE) and undersampling of majority class have been used for handling imbalanced data.

- **Ensemble Learning:** Random Forest was combined with other models in order to get a stronger predictive system. Among the usual method applied to enhance more performance are stacking and blending.

- By applying these techniques, we aimed to improve the performance and robustness of the Random Forest model in stock price prediction, ensuring that the model is well-suited for practical applications in financial mar- kets.

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4. Results and Discussion

In total there are four categories of machine learning models and they are: Random Forest classifier, Logistic Regression. If the algorithms employed were Support Vector Machine (SVM), and Gradient Descent—on. the bur- den of forecasting of the stock price. The models were evaluated a comparison of several performance indicators, and with accuracy as primary metric. As well, Mean Absolute Error (MAE), In order to evaluate each coeffi- cient of determination (R2), root mean square error (RMSE) was computed. Average Squared Error, ASE, Root Mean Squared Error provide extra level of protection for another model's performance. These metrics kept a path to the comprehen-sive goals that were articulated at HRM's beginning: talistic approach towards dissecting the constraints of the variety of the models within reference to the predictive vogue.

Table 1. Performance Metrics for Different Models

Model	Accuracy	MAE	MSE	RMSE
Random Forest	86.24%	1.45	2.11	1.45
Logistic Regression	85.71%	1.53	2.23	1.49
SVM	84.65%	1.64	2.42	1.55
Gradient Descent	78.30%	2.35	4.11	2.03

4.1 Model Performance

Table 1 summarizes the performance metrics (Accuracy, MAE, MSE, RMSE) for each model.

- **Random Forest** the Random Forest model demonstrated best accuracy at 86.24%, positioning it as the most effective model in this comparative study. In addition to its high accuracy, it got a MAE of 1.45, an MSE of 2.11, and an RMSE of 1.45, showing its robust ability to generalize to unseen data while maintaining low prediction errors. These results can be attributed to Random Forest's ensemble learning approach, which combines the outputs of multiple decision trees to reduce overfitting and improve predictive power.

- **The Random Forest** model's strength lies in its ability to capture complex, non-linear relationships between stock price features, such as the open, close, and volume, while providing a high degree of interpretability through feature importance analysis. This analysis revealed that the features most influential to the model's predictions included the stock's opening price and trading volume, confirming that these indicators play a significant role in the model's performance. The model's capacity to handle large datasets and its resilience to noisy data, which is often present in stock markets, further explain its superior performance.

- **Logistic Regression** Despite being a relatively simpler model, Logistic Regression

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achieved a close second with an accuracy of 85.71 Logistic Regression is typically used for binary classification tasks, but in this study, it was adapted for re- gression to predict continuous stock price values. Its high performance can be attributed to the dominance of linear trends in the stock data; however, it struggled to capture more complex dependencies, which limited its overall pre- dictive power. Nevertheless, its simplicity and ease of interpretation make it a strong candidate when transparency and linear relationships dominate the problem domain.

- **Support Vector Machine (SVM)** The SVM model, using a linear kernel, achieved an accuracy of 84.65 SVMs are characterized by their high dimensional data hawk-eye and the capability of successfully construct- ing a clear platform of decision-making but the restricted employments of the kernel linear programs did not allow the identification of non-linearity features of the stock price data set. Thus, using a non-linear kernel say a RBF or a polynomial kernel might increase its performance level as the data collected in stock price cannot be reasonably expected to live in a linear space. Further optimization of the hyperparameters, such as the penalty parameter C and the kernel coefficient γ, might also lead to enhanced results. Despite its slightly lower performance, SVM remains a powerful tool in regression tasks when more sophisticated kernel functions are applied.

- **Gradient Descent** the Gradient Descent performs the worst with a minimum accuracy of 78.30%, MAE of 2.35, MSE of 4.11 and RMSE of 2.03. Gradient Descent is the core optimization algorithm used in many of the machine learning models, but here, in this context, the linear nature in itself posed a limitation on its ability to model these non-linear dependencies critical for stock price prediction. Low accuracy and error score this mean that in the simple form of optimization with the strategy based on gradient, for a problem with high-volatility and with nonlinear correlations like stock price prediction. Though Gradient Descent is highly effective in reducing loss function, it falls very short of expectation without other advanced modeling when applied on stock data with intricate trends.

5. Discussion

The comparison between the performance of the four models demonstrates which literally reveals that ensemble methods, particularly Random Forest, are highly have proved to be a viable solution for stock price prediction. The better performance of Random Forest can be attributed to its ability to combine conventional methods of feature selection with the construct of using a randomly selected subset of the data to build base models. multiple decision trees, which allows it to model both simple and complex decision processes that occur in organisations. as well as regarding features and variations of the data pattern. This model averages multiple predictions to be able to generalize and perform well on unknown data.

Logistic Regression, despite being simpler and more interpretable, performed impressively well in terms of forecasting stock prices, so linear trends appear to still dominate historical stock data used in the study, although its slight inferiority in comparison with Random Forest tells that more complex models are needed in order to handle the really complex movements of stock prices Fig 1.

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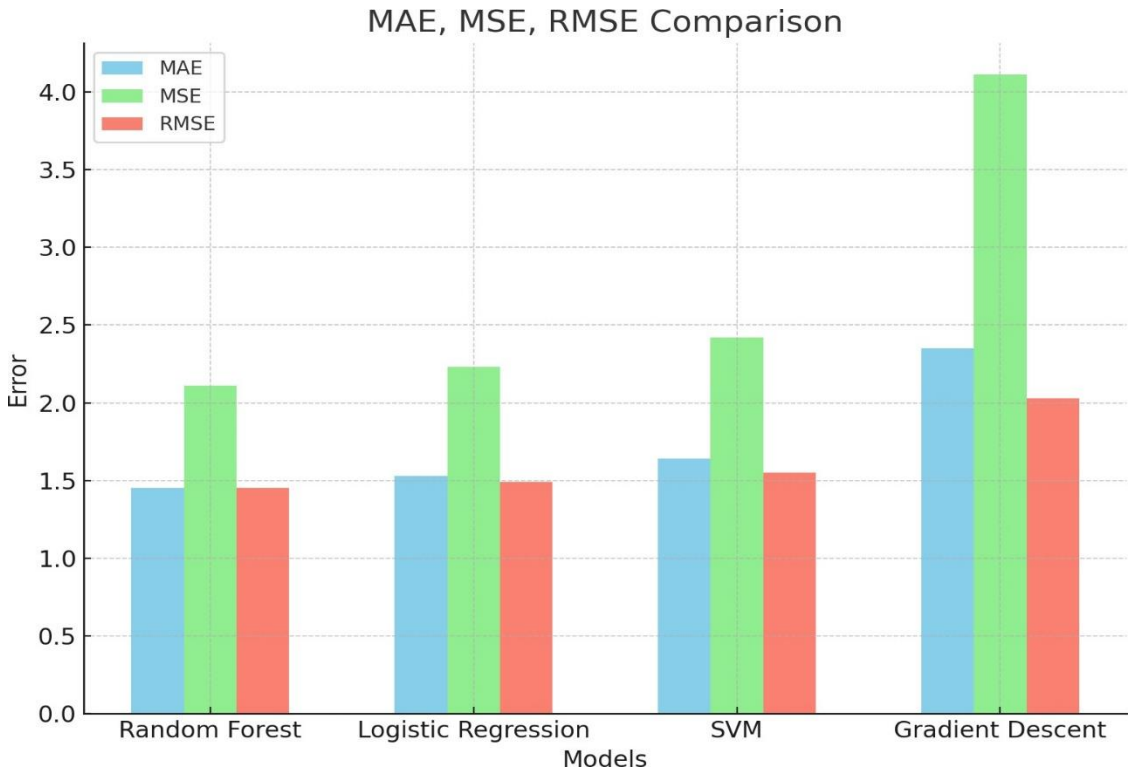


Figure 1. MAE, MSE, RMSE Comparison Chart

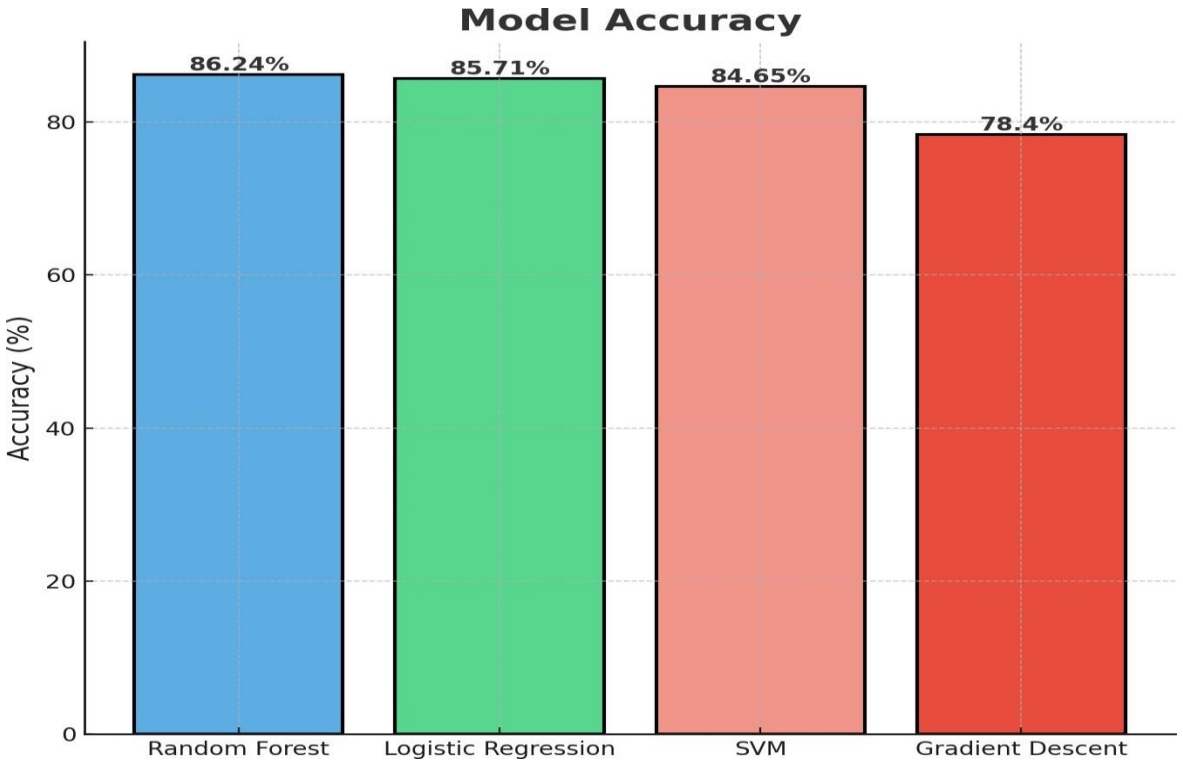


Figure 2. Model Accuracy Chart

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The SVM model did well but still had some potential for further improvement. If it was fitted using a linear kernel, its effectiveness was in capturing linear associations rather than non-linear ones. More importantly, the use of non-linear kernels or higher-order optimization of hyperparameters might improve its accuracy and reduce errorin predictions.

Gradient Descent is a very simple optimization technique, but it's the worst approach mostly because of its linearity. The model performed terribly, and by comparison, this kind of approach only emphasizes more advanced modeling methodologies capable of performing general forms of non-linear relationships often associated with stock price data.

The Predicted vs. Actual plot for Random Forest (Figure 2) further illustrates the model's accuracy in fore- casting stock prices. Most of the predictions closely match the actual values, indicating that Random Forest is a reliable model for this task.

6. Conclusion

A comprehensive comparative analysis between four machine-learning models used for predicting stock price have been presented in this research. It is noted that Random Forest was the best model here, with Logistic Regression and SVM just behind. Gradient Descent Another algorithm that was used in the analysis: While it is a powerful tool to optimize a function, it does not perform as well when it comes to modeling non-linearities in stock price trends. Finally, we also revisit some potential avenues for improving Random Forest (hyperparameter tuning, feature engineering, etc.)

Implication of the findings of this study the study provides the following conclusions: comparative advantages and disadvantages of various models of stock prices. prediction, which gives the basis for further investigations. and its use in the financial market and associated fields, including practice.

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